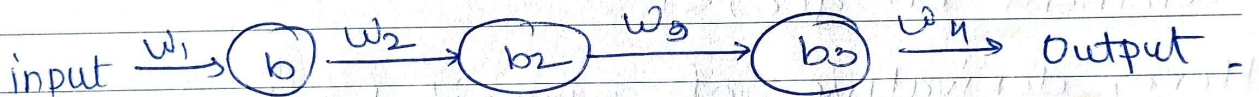


The Vanishing Gradient problem

Vanishing Gradient: It's a state where derivatives of sigmoid function becomes so low that it becomes impossible to update older weight with new one because after multiple epochs and derivation, both values becomes same. In that case, it becomes harder to achieve minima.

Given:



Output $(y) = f_w(x) = w_4 g(z_3)$, where

$$g(z) = \frac{1}{1 + e^{-z}}$$

$$z_i = w_i g(z_{i-1}) + b_i$$

$$C = \frac{1}{2} [f_w(x) - a(x)]^2$$

a) To find the derivative of C , w.r.t. b_1 we need to apply chain rule.

$$\therefore \frac{\partial C}{\partial b_1} = \frac{\partial C}{\partial f_w} \cdot \frac{\partial f_w}{\partial z_3} \cdot \frac{\partial z_3}{\partial z_2} \cdot \frac{\partial z_2}{\partial z_1} \cdot \frac{\partial z_1}{\partial b_1}$$

Let's compute each part:

$$\frac{\partial C}{\partial f_w} = f_3(x) - a(x)$$

$$\frac{\partial f_w}{\partial z_3} = \frac{\partial (\omega_4 g(z_3))}{\partial z_3} = \omega_4 g'(z_3) \text{ , Here, } g'(z_3) = \frac{\partial g(z_3)}{\partial z_3}$$

$$\frac{\partial z_3}{\partial z_2} = \frac{\partial (\omega_3 g(z_2))}{\partial z_2} = \omega_3 g'(z_2)$$

$$\frac{\partial z_2}{\partial z_1} = \omega_2 g'(z_1)$$

$$\frac{\partial z_1}{\partial b_1} = \frac{\partial (\omega_1 x + b_1)}{\partial b_1} = 1$$

$$\therefore \frac{\partial C}{\partial b_1} = (f_w(x) - a(x)) \cdot \omega_4 g'(z_3) \cdot \omega_3 g'(z_2) \cdot \omega_2 g'(z_1)$$

Comment on $g'(z) < 1$:

Here, derivative of sigmoid function $g(z)$ is a $g'(z) = g(z)(1-g(z))$, which satisfies condition $g'(z) < 1$ because $g(z)$ is always in between 0 to 1

And if we do the multiplications of these weights multiple times, $g'(z)$ will become very small which results in Vanishing Gradient Problem where $\frac{\partial C}{\partial b_1} \approx 0$, which makes difficult to update the weights.

b) condⁿ $|\omega g'(\omega z + b)| \geq 1$ Requires $|\omega| \geq 4$.

Here, maximum value of the $g'(z)$ occurs when $z = 0$, which is substituting value $z = 0$ in $g'(z)$

$$g'(0) = g(0)(1-g(0)) = \frac{1}{1+e^0} \left(1 - \frac{1}{1+e^0}\right) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = 0.25$$



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$$\text{Now, } |\omega| \cdot g'(\omega z + b) \geq 1$$

$$|\omega| \cdot \frac{1}{4} \geq 1 \Rightarrow |\omega| \geq 4$$

From this, we can conclude that only if $|\omega| \geq 4$,
equations value can exceed 1.

$$\text{c) Here, } |\omega| g'(\omega z + b) \geq 1$$

$$\text{let's } u = \omega z + b$$

$$\therefore |\omega| \cdot g'(u) \geq 1 \Rightarrow g'(u) \geq \frac{1}{|\omega|}$$

$$\text{Now, } g'(u) = g(u)(1 - g(u))$$

$$\text{let's } g(u) = y$$

$$\therefore y(1 - y) \geq \frac{1}{|\omega|}$$

$$\therefore y^2 - y + \frac{1}{|\omega|} \leq 0$$

lets solve this quadratic eqⁿ, and find
it's roots.

$$a = 1, b = -1, c = \frac{1}{|\omega|}$$

$$\therefore y \in \frac{1 \pm \sqrt{1 - 4/|\omega|}}{2}$$

This two roots gives us the range for y
for which it's negative to zero.

let's substitute the back $u = \omega z + b$

$\therefore y = g(u)$, let's invert the sigmoid to find u :

$$u = \ln\left(\frac{y}{1-y}\right)$$

The width of u is twice the value at $y = \frac{1 + \sqrt{1 - 4/|w|}}{2}$

$$\therefore u = \ln\left(\frac{\frac{1 + \sqrt{1 - 4/|w|}}{2}}{1 - \left(\frac{1 + \sqrt{1 - 4/|w|}}{2}\right)}\right)$$

Let's multiply by 2,

$$\Delta u = 2 \ln\left(\frac{1 + \sqrt{1 - 4/|w|}}{1 - \sqrt{1 - 4/|w|}}\right)$$

Now, since $u = wz + b$, so corresponding range for z is

$$\Delta z = \frac{\Delta u}{|w|} = \frac{2}{|w|} \ln\left(\frac{1 + \sqrt{1 - 4/|w|}}{1 - \sqrt{1 - 4/|w|}}\right)$$

$$\therefore \Delta z = \frac{2}{|w|} \ln\left(\frac{|w|(1 + \sqrt{1 - 4/|w|})}{2} - 1\right)$$

$\therefore$ by solving with the rule $\frac{a+b}{a-b}$

$\therefore \Delta z$ which represents range for z i.e. $\{z\}$, matches the given eqⁿ.

d) $|w| \approx 6.9$

$$\therefore \Delta z = \frac{2}{6.9} \ln\left(\frac{6.9(1 + \sqrt{1 - 4/6.9})}{2} - 1\right) = 6.447742$$

$$\Delta z \approx 0.45$$

$\therefore$ we can say that Δz is highest at $|w| = 6.9$